REBASE: Device-Cloud Experience Coherence for GUI Agents Across App Updates

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Paper version: arXiv 2609.32130v1 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *An app update can make these paths partly wrong without making them unusable: the failure is silent.*

Cached on-device **experience** goes stale after an update: **10.1 points** below no experience.

Motivation: three observations

1. **Silent failure.** A broken step is still walkable in 62.2% of runs.
2. **Late detection.** The mismatch shows 3 steps after acting.
3. **Unused signals.** The cloud may see the new version first.

Evaluation setup

AndroidWorld · 9 apps | 45 units · 5 tasks each | 6 visible, 3 blind pairs | Qwen3-VL-8B INT4 | Gemini 3.1 Pro cloud | K = 5 episodes | 3 seeds

- **All online:** the evaluator reads app state.
- **Partial hold-out:** 3 blind pairs hide the new version from the cloud.

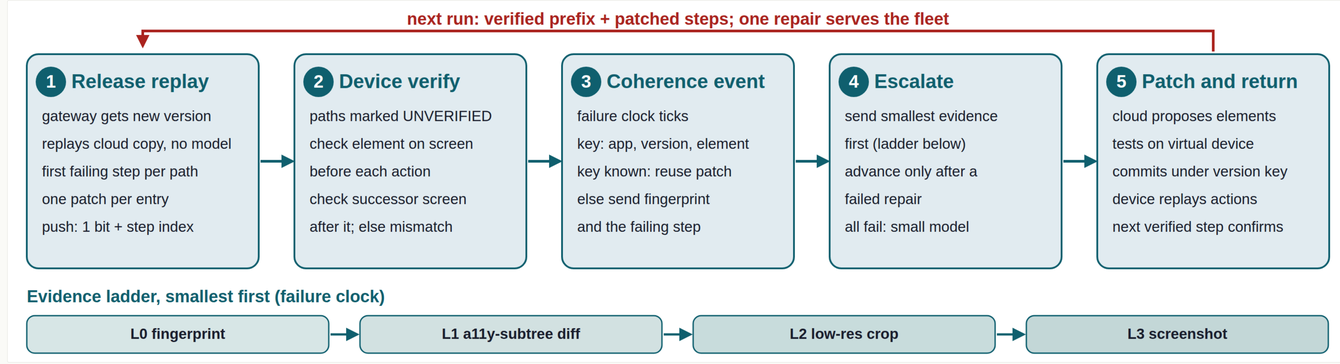
Our read (CUA Radar)

68	★★	online
score of 100	band	evidence level

- **Reports:** online eval; std over 3 seeds; takeover and upload baselines; matched K; cost; ablation. **Partial:** blind pairs.
- **Flags:** none. Code: not stated.
- Pattern C19. Authors' numbers, not rerun.

Method: two clocks over one experience cache

- **Sees / acts:** accessibility tree in; recorded steps replayed on fingerprint-matched elements.
- **Update clock:** cloud replays its copy on the new version; device verifies each step.
- **Failure clock:** smallest evidence first; cloud returns a tested patch under a version key.



Our redraw (not the paper's figure): update clock, failure clock and the evidence ladder.

$$\nu_k(p, v') = \mathbb{I}\left[f_k \simeq S_k(v')\right] \mathbb{I}\left[S_{k+1}(v') \simeq S^{\star}_{k+1}\right]$$

Step verification: element present, successor screen matches

Key results: online Succ@K (%), AndroidWorld

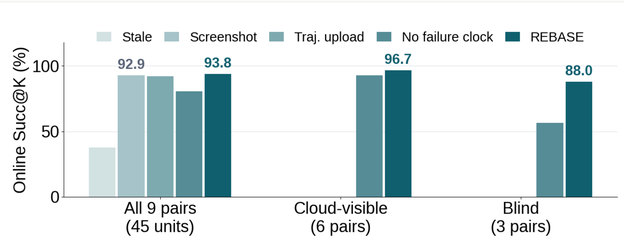


Table I and Sec. V, redrawn: K = 5, 3 seeds. Stale = unrepaired; no experience (47.7) not charted.

+56.1 pts over stale experience (95% CI [51.6, 60.6])

SO WHAT → *Screenshot-to-cloud matches Succ@K because the cloud completes every broken episode.*

Ablation: Succ@K (%), nine pairs

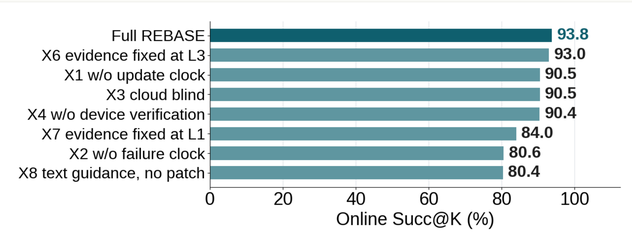


Table I lower block, redrawn: one component removed or fixed at a time.

SO WHAT → *Text guidance (X8) and no failure clock (X2) cost most; L3 matches but at 14.4× the uplink. With the cloud blind (X3), success stays within 3.3 pts, but recovery in one episode drops from 83.7% to 76.3%. Without device verification (X4), stale steps per episode rise from 0.1 to 3.7. Uplink 50 to 2 Mbps: success within 2.7 pts.*

Headline Numbers

93.8%
Succ@K, K = 5
stale 37.6%; none 47.7%

- **708×** fewer bytes vs upload
- **47.3×** fewer cloud calls
- **2.5×** less energy vs none

The paper's own controls

CONTROL (ONLINE)	REBASE	COMPARED WITH
Re-explore on failure	93.8	69.3
Blind pairs (3)	88.0	56.4 w/o failure clock
Recovery in episode 1	91.1% vis. / 68.9% blind	re-explore 28.1%
Device J per episode	702	539 shot.; 152 cloud-only

Takeaway

Verify on update, escalate on failure: stale experience is restored to fresh-experience success at 708× fewer bytes than screenshot upload, on one Android testbed.

80.0× fewer re-derivations | no code stated

Scan to read the full paper

arxiv.org/abs/2609.32130v1